

SE3070 -Case Studies in Software Engineering

Assignment 01

Case Study 2: Smart Disaster Early-Warning and Emergency Coordination System for Sri Lanka



Group ID – Group_026

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1. Introduction

Sri Lanka faces recurring natural hazards including monsoon flooding, landslides in the central highlands, cyclones along the coast, and occasional drought. Response effectiveness today depends heavily on how quickly warnings reach the public and how efficiently relief resources are coordinated across the Disaster Management Centre (DMC), district secretariats, police, armed forces and NGOs. Coordination currently relies on phone calls, radio broadcasts, WhatsApp groups and paper-based situation reports, which leads to duplicated effort, delayed evacuations and misallocated relief supplies.

This report presents the Phase 1 design of a unified digital platform for the DMC: a citizen mobile application and web portal linked to a central hazard-monitoring dashboard used by DMC and district officers. The design addresses hazard warning broadcast, citizen/volunteer ground reporting with officer verification, multi-agency shelter and resource coordination, and post-event statistical reporting.

Section 2 presents the group deliverables -a high-level use case diagram covering four major business use cases, and a class diagram derived from those use cases. Sections 3–5 present each member's individual deliverables for one selected use case: a detailed use case scenario, a sequence diagram, a storyboard, and low-and high-fidelity wireframes.

2. Group Component -Functional Design

2.1 High-Level Use Case Diagram

Four major business use cases were identified directly from the four core capabilities described in the case study: broadcasting hazard warnings, capturing and verifying citizen ground reports, coordinating multi-agency shelter and resource response, and analyzing post-event data. Login, logout, and administrative-privilege functions are intentionally excluded, as instructed, since they are not domain-specific.

Smart Disaster Early-Warning and Emergency Coordination System (Sri Lanka)



Figure 1: High-level use case diagram -four major business use cases and seven actors.

Use case summaries

- UC-01 Issue Hazard Warning -DMC/Duty Officer selects a hazard type, severity and target scope (district or river basin) and broadcasts push, SMS and audible alerts to citizens in that scope.
- UC-02 Submit and Verify Hazard Report -Citizens and community volunteers submit ground reports (photo + GPS + description); a Duty Officer reviews and verifies each report before it can influence any official warning (<<extend>> relationship into UC-01).
- UC-03 Coordinate Shelter and Resource Allocation -District Officers register shelters, track occupancy, dispatch rescue teams and log relief supplies that may be owned by government, armed-forces or NGO/donor organisations, while retaining a single combined operational picture.

- UC-04 Generate Post-Event Analysis Report -DMC officers and donor/NGO representatives generate statistical reports covering alert timelines, citizens reached, shelter occupancy over time and resource distribution by district.

2.2 Class Diagram

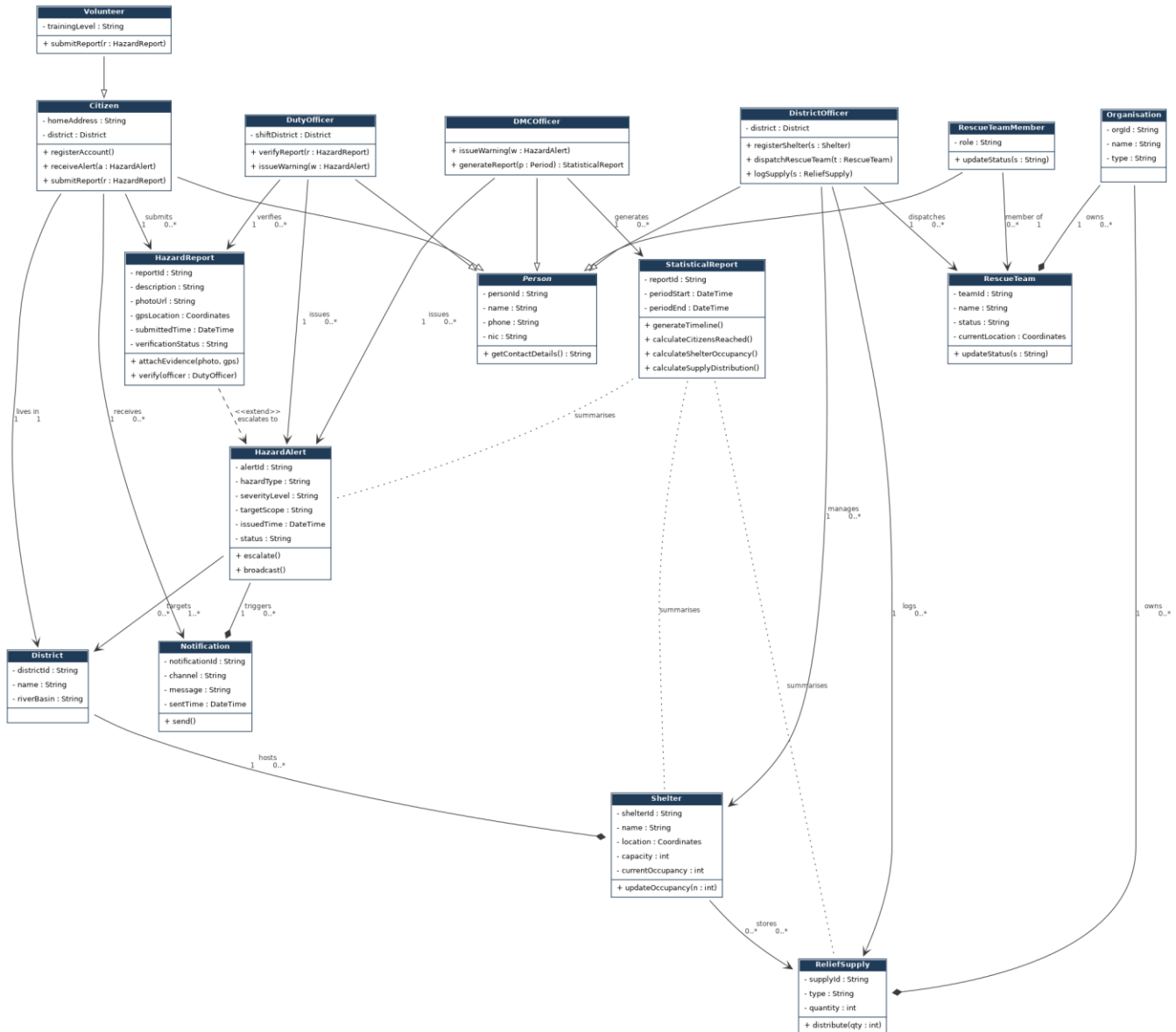


Figure 2: Class diagram covering all four business use cases, with inheritance, association, aggregation and dependency

3. Individual Component -W.W.T.S. Fernando IT23179530 : Issue Hazard Warning

3.1 Use Case Scenario

Use Case ID	1
Use Case Name	Issue Hazard Warning
Actors	Primary: Duty Officer , DMC Officer. Secondary: Citizen, Notification Service.
Description	Allows an authorized officer to broadcast a location specific hazard warning
Trigger	The officer identifies, from the monitoring dashboard, verified ground reports, or external hazard data, that a hazard warrants public notification.
Preconditions	• Officer is authenticated on the DMC dashboard with warning-issuance privileges. • The affected districts or river basins are already registered in the system.
Postconditions	• A Hazard Alert record exists with status BROADCAST. • Notifications have been dispatched to all citizens registered within the target scope. • The alert is available for later inclusion in a statistical report .
Main Flow	1. Duty Officer opens the “Issue Hazard Warning” screen on the dashboard. 2. Officer selects the hazard type (Flood, Landslide, Cyclone, Drought). 3. Officer sets the severity level (Low / Medium / High / Severe). 4. Officer defines the target scope by selecting affected districts or river basins. 5. System validates the target scope against the registered District / River Basin list. 6. Officer reviews the auto-generated broadcast preview message. 7. Officer confirms and submits the broadcast. 8. System creates the HazardAlert record (status DRAFT) and escalates it to BROADCAST. 9. System dispatches push, SMS and audible notifications to all citizens within the target scope. 10. System records delivery confirmations and displays a summary to the officer. 11. Use case ends successfully.
Alternative Flows	A1 -Escalation from a verified citizen report: A1.1 A citizen/volunteer hazard report has already been verified . A1.2 Officer opens the verified report and selects “Escalate to Warning”. A1.3 Hazard type and approximate location are pre-populated from the report. A1.4 Flow resumes at Main Flow step 3.
Exception Flows	E1 -Invalid target scope: E1.1 At step 5, the system finds no matching district / river basin record. E1.2 System displays a validation error and highlights the field; officer corrects the scope and resumes at step 4. E2 -Notification channel unreachable: E2.1 At step 9, the primary channel (push) fails for some devices due to a network outage. E2.2 System automatically retries via the SMS gateway and queues the audible-alert trigger for reconnection. E2.3 Officer sees a partial-delivery warning in the summary at step 10.

3.2 Sequence Diagram

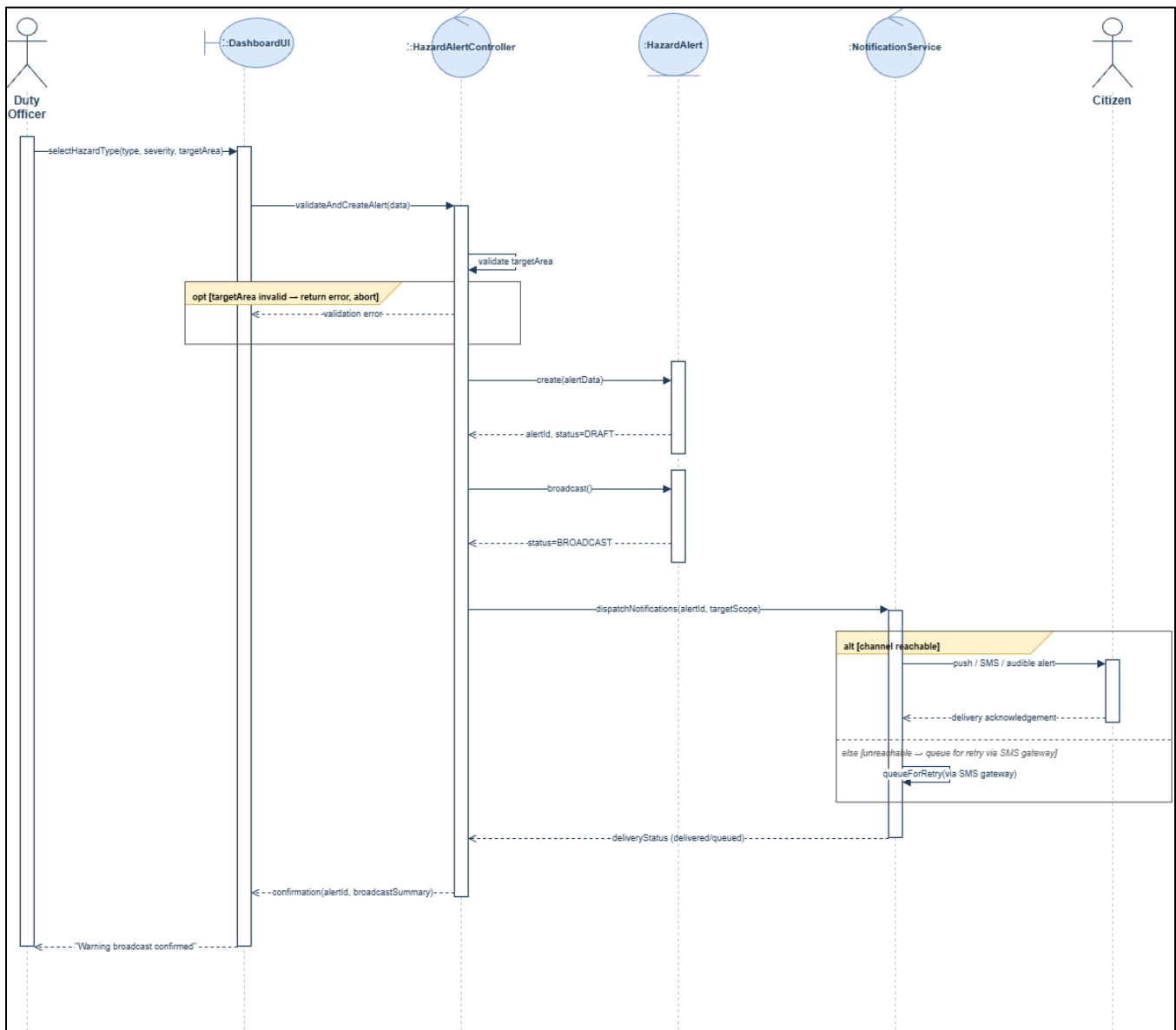


Figure 3: Sequence diagram -Issue Hazard Warning.

3.3 Storyboard



Figure 5: Storyboard – Issue Hazard Warning (Duty Officer)

3.4 Low-Fidelity Wireframe

Issue Hazard Warning

Hazard Type

Flood ▼

Severity Level

Low

Medium

High

Severe

Target Scope (District / River Basin)

Broadcast Preview

Cancel

Confirm & Broadcast

Figure 6: Low-Fidelity Wireframe – Issue Hazard Warning

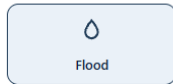
3.5 High-Fidelity Wireframe



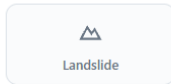
Issue Hazard Warning

Broadcast a location specific hazard warning to citizens in the affected districts. Alerts are delivered by push notification, SMS and audible alert simultaneously.

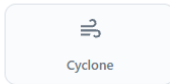
1 Hazard Type



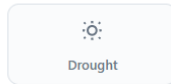
Flood



Landslide



Cyclone



Drought

Severity Level

LOW

MEDIUM

HIGH

SEVERE

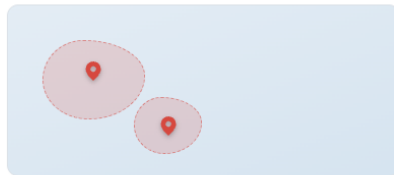
2 Target Scope

☒ Colombo District

☒ Gampaha District

☐ Kalutara District

☐ Kelani River Basin



Broadcast Preview



Flood Warning : SEVERE

Heavy rainfall expected in Colombo & Gampaha. Risk of flooding. Move to higher ground and follow official guidance.

DMC Alert - now



Push notification

Ready



SMS gateway

Ready



Audible alert

Ready

48,200 citizens in target scope

Status: DRAFT

Cancel



Confirm & Broadcast

4. Individual Component -W.A.M.N. Fernando (IT23562660) (Submit and Verify Hazard Report)

4.1 Use Case Scenario

Use Case ID	UC-02
Use Case Name	Submit and Verify Hazard Report
Actors	Primary: Citizen, Community Volunteer. Secondary: Duty Officer.
Description	Allows citizens and community volunteers to submit ground-level hazard reports (photo, GPS location, description) which a Duty Officer must review and verify before they can influence any official warning level.
Trigger	A citizen or volunteer observes a ground-level hazard indicator, such as a rising river level, a blocked road, or a landslide crack.
Preconditions	• Citizen/volunteer has a registered account (home address and district on file) or has granted location permission for guest use. • Device has a functioning camera and, ideally, GPS.
Postconditions	• A HazardReport exists with verificationStatus = CONFIRMED or DISMISSED. • If CONFIRMED, the report is available to be escalated into a HazardAlert (UC-01, Alternative Flow A1).
Main Flow	1. Citizen/Volunteer opens the mobile app and selects “Report a Hazard”. 2. App activates the camera; the user captures a photo of the hazard. 3. App auto-tags the current GPS location. 4. User enters a short text description and selects a hazard type from quick-select chips. 5. User submits the report. 6. System creates a HazardReport with status PENDING, storing the photo, GPS location and description. 7. System notifies the on-duty officer's dashboard of the new pending report. 8. Duty Officer opens the report and reviews the photo, location and description. 9. Officer verifies the report and marks it CONFIRMED. 10. System updates the report status and notifies the submitting citizen. 11. Use case ends successfully.
Alternative Flows	A1 -Report rejected as a false alarm: A1.1 At step 9, the officer determines the report is inaccurate or a duplicate. A1.2 Officer marks the report DISMISSED with a reason code. A1.3 System closes the report without affecting the current warning level; the submitting citizen is notified politely.
Exception Flows	E1 -GPS/network unavailable at submission: E1.1 At step 3 or 5, the device has no GPS fix or no network connectivity. E1.2 App caches the report (photo, description, last-known location) locally. E1.3 When connectivity is restored, the app automatically syncs the queued report and resumes at step 6. E2 -Duplicate report detected: E2.1 At step 6, the system detects another pending report within a small radius and time window. E2.2 System links the reports as a cluster and flags this for the officer, rather than treating them as unrelated duplicates.

4.2 Sequence Diagram

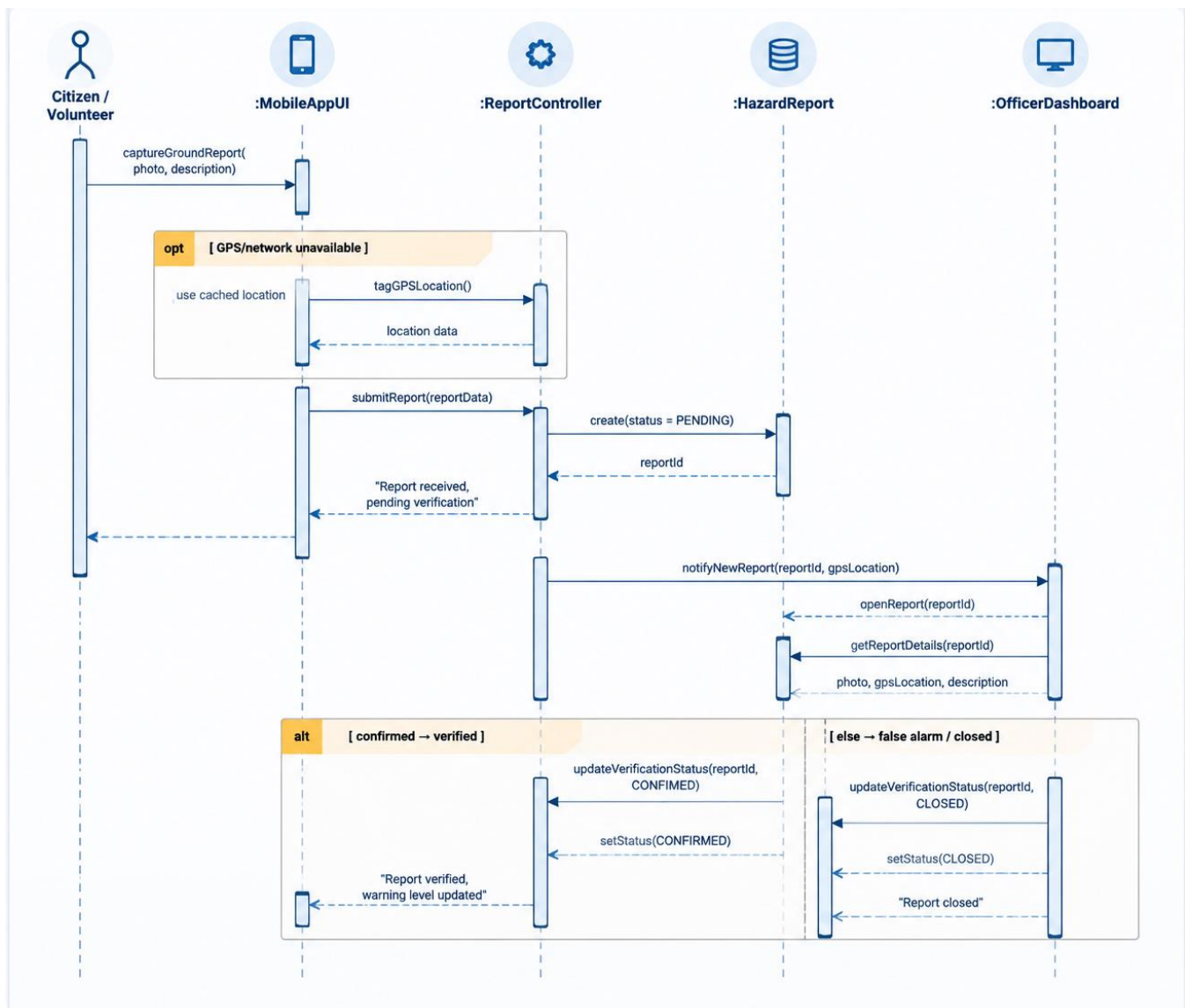


Figure 3: Sequence diagram -Submit and Verify Hazard Report.

4.3 Storyboard

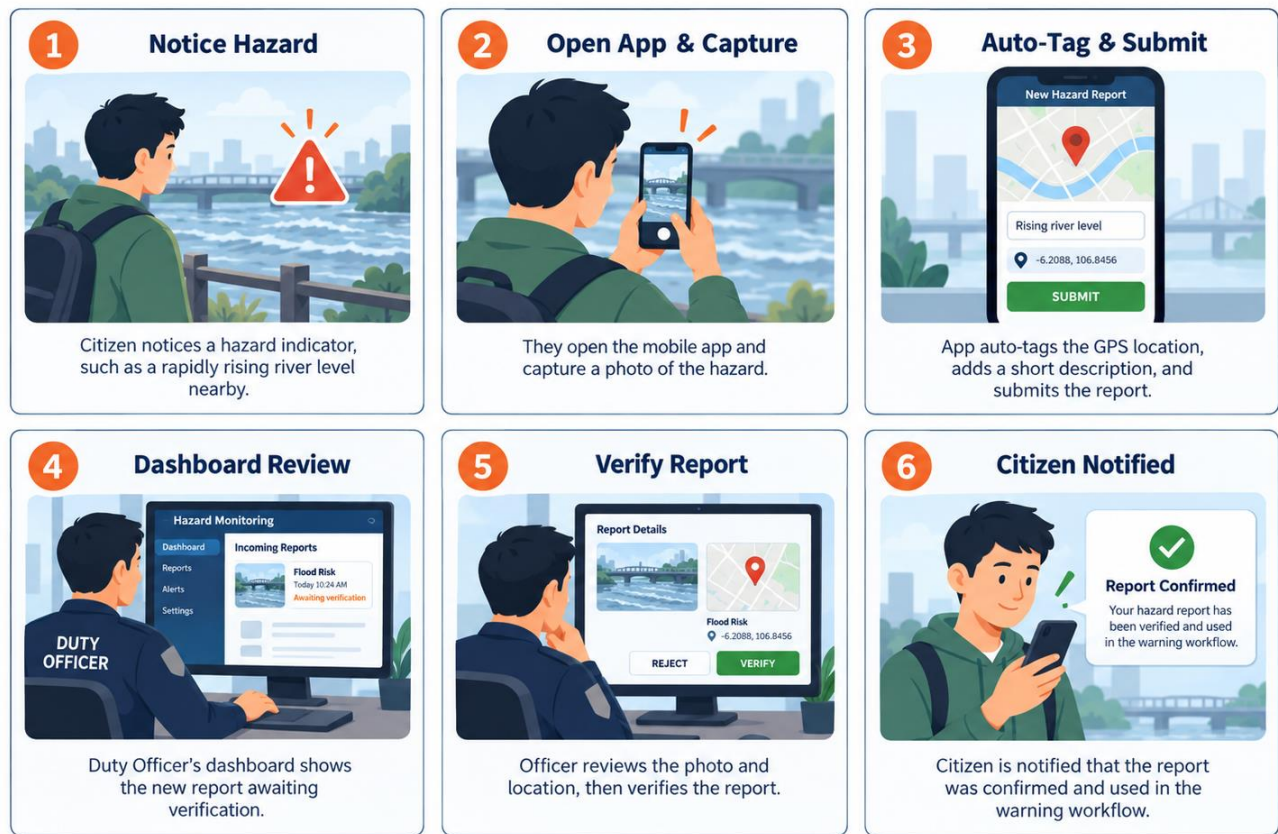
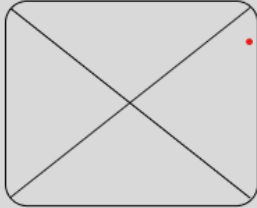


Figure 5: Storyboard – Submit and Verify Hazard Report

Figure 4: Storyboard -Citizen submits and Duty Officer verifies a hazard report.

4.4 Low-Fidelity Wireframe

Report a Hazard



Location: auto detected

Description

Hazard Type

Flood

Landslide

Road Block

Submit Report

Figure 5: Low-fidelity wireframe -Submit Hazard Report screen.

4.5 High-Fidelity Wireframe

←

Report Hazard





Tap to take photo



79.797.79.79797 ' E

Location detected

Change

Description

Write a short description...

0/200

Hazard Type



Flood



Landslide



Road Block

 Submit Report

Figure 6: High-fidelity wireframe -Submit Hazard Report screen.

5. Individual Component -S.R.L. Bandara (IT23580558) (Coordinate Shelter and Resource Allocation)

5.1 Use Case Scenario

Use Case ID	UC-03
Use Case Name	Coordinate Shelter and Resource Allocation
Actors	Primary: District Officer. Secondary: Rescue Team Lead, Organisation (Government / Armed Forces / NGO / Private Donor).
Description	Allows a District Officer to track shelter occupancy against capacity, dispatch rescue teams, and log relief-supply distribution, while resources owned by different organisations remain individually attributable and DMC officers retain one combined operational picture.
Trigger	An official hazard warning has been escalated for the district (UC-01) and response coordination is required.
Preconditions	- Shelters, rescue teams and relief-supply stock for the district are already registered (or are registered as part of this flow). - District Officer is authenticated with resource-management privileges for that district.
Postconditions	- Shelter occupancy figures are current. - Rescue-team dispatch status is recorded and visible to DMC. - Relief-supply distribution is logged against a specific shelter and owning organisation. - DMC officers have a combined operational picture spanning all owning organisations.
Main Flow	- District Officer opens the Shelter & Resource Coordination Dashboard. - Officer updates the current occupancy of an affected shelter. - System stores the new occupancy value against the Shelter record. - Officer selects an incident location and dispatches the nearest available Rescue Team. - System sets the team's status to DISPATCHED and records the target location. - Rescue team acknowledges the dispatch from a field device; status updates to ON SITE on arrival. - Officer logs relief supplies (type and quantity) distributed to a shelter, noting the owning organisation. - System records the ReliefSupply entry, linked to the Shelter and Organisation. - Dashboard refreshes to show a combined operational picture across all owning organisations. - Use case ends successfully.
Alternative Flows	A1 -Shelter nearing capacity: A1.1 At step 3, updated occupancy reaches or exceeds 90% of capacity. A1.2 System flags the shelter and suggests the nearest alternate shelter with spare capacity. A1.3 Officer may redirect further evacuees to the alternate shelter; flow resumes at step 2 for the new shelter.
Exception Flows	E1 -Rescue team unresponsive: E1.1 At step 6, the dispatched team does not acknowledge within the configured timeout. E1.2 System escalates the dispatch and prompts the officer to reassign the task to the next available team. E1.3 Flow resumes at step 4 with the reassigned team. E2 -Resource owned by another organisation: E2.1 At step 7, the supply belongs to an NGO or armed-forces organisation rather than the DMC. E2.2 System still records the distribution against that organisation's ownership, retaining accountability while including it in the DMC's combined operational view.

5.2 Sequence Diagram

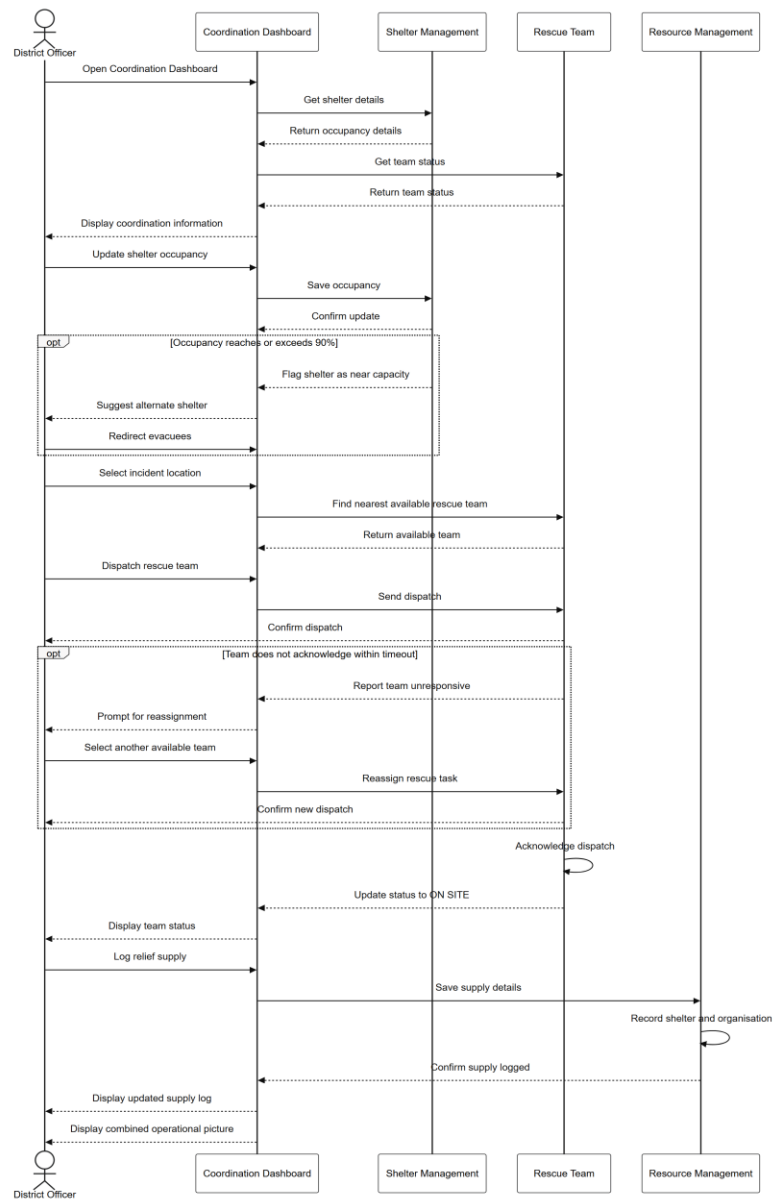


Figure 3: Sequence diagram -Coordinate Shelter and Resource Allocation.

5.3 Storyboard

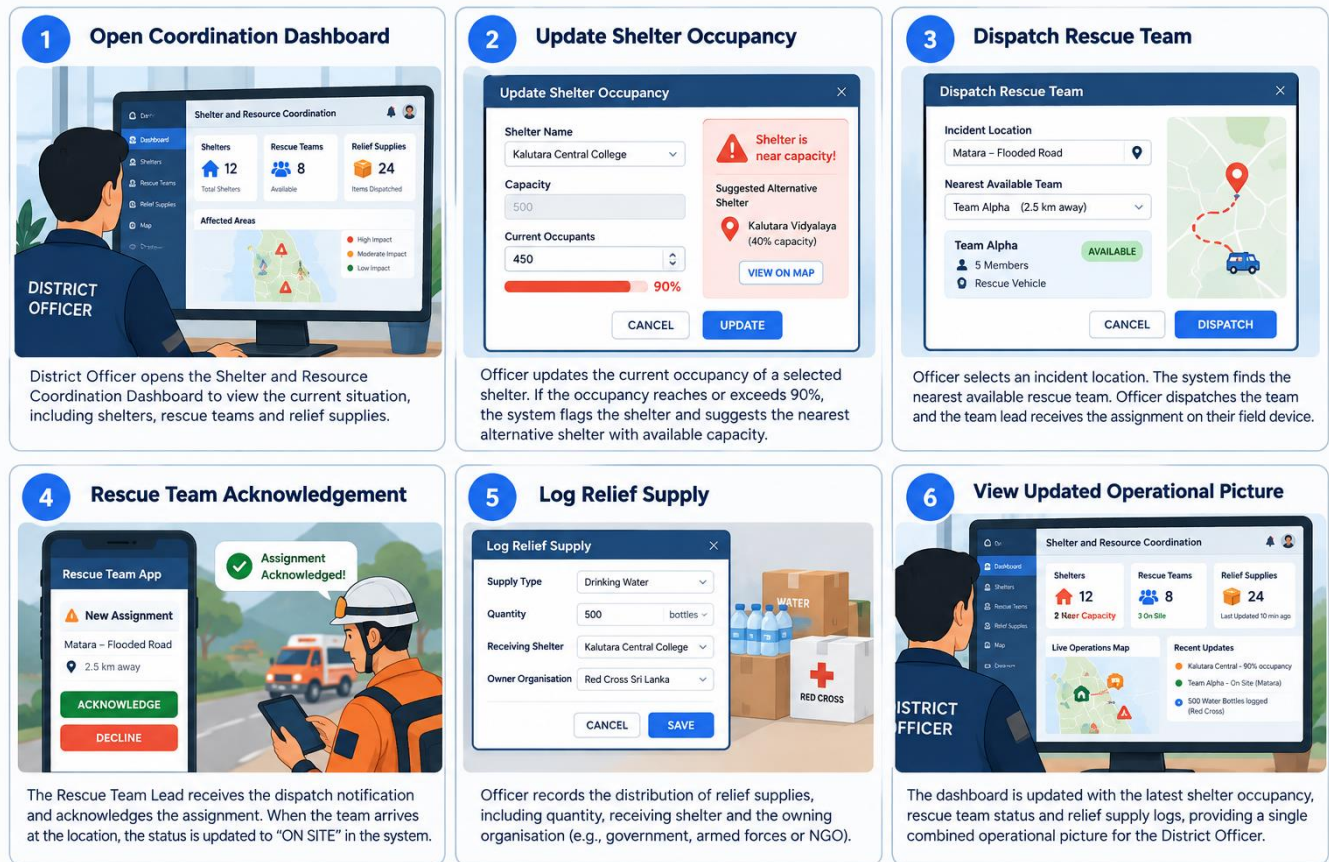


Figure 4: Storyboard -District Officer coordinates shelters, teams and supplies.

5.4 Low-Fidelity Wireframe

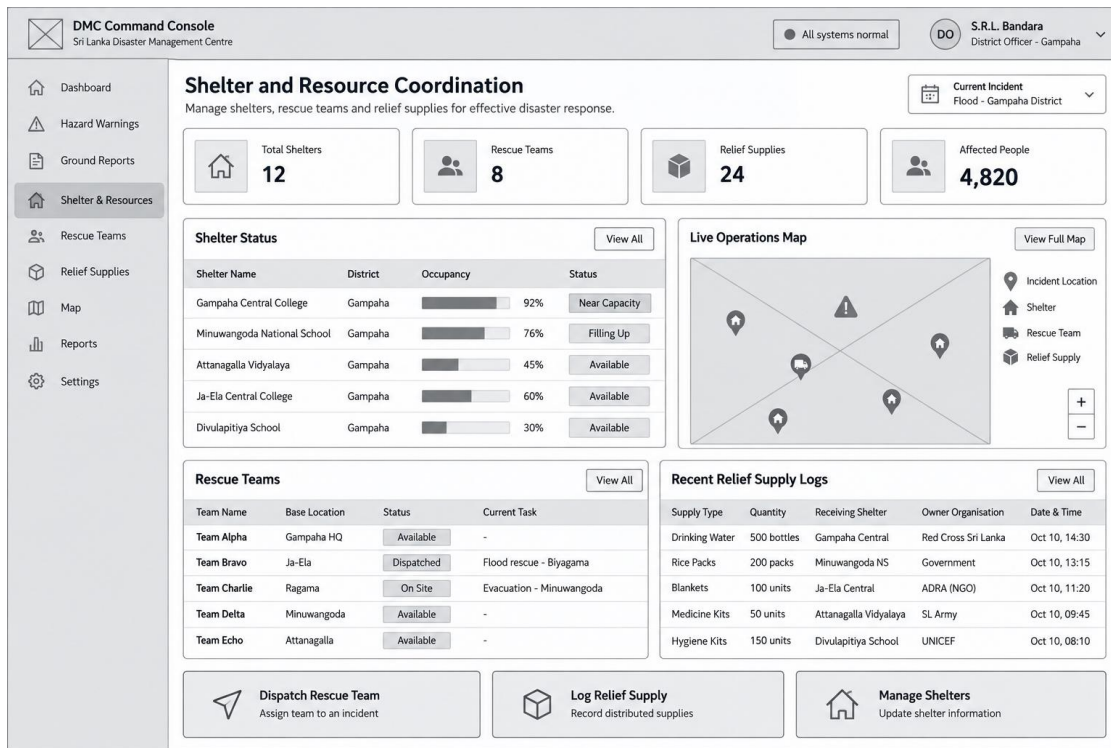


Figure 5: Low-fidelity wireframe -Shelter and Resource Dashboard.

5.5 High-Fidelity Wireframe

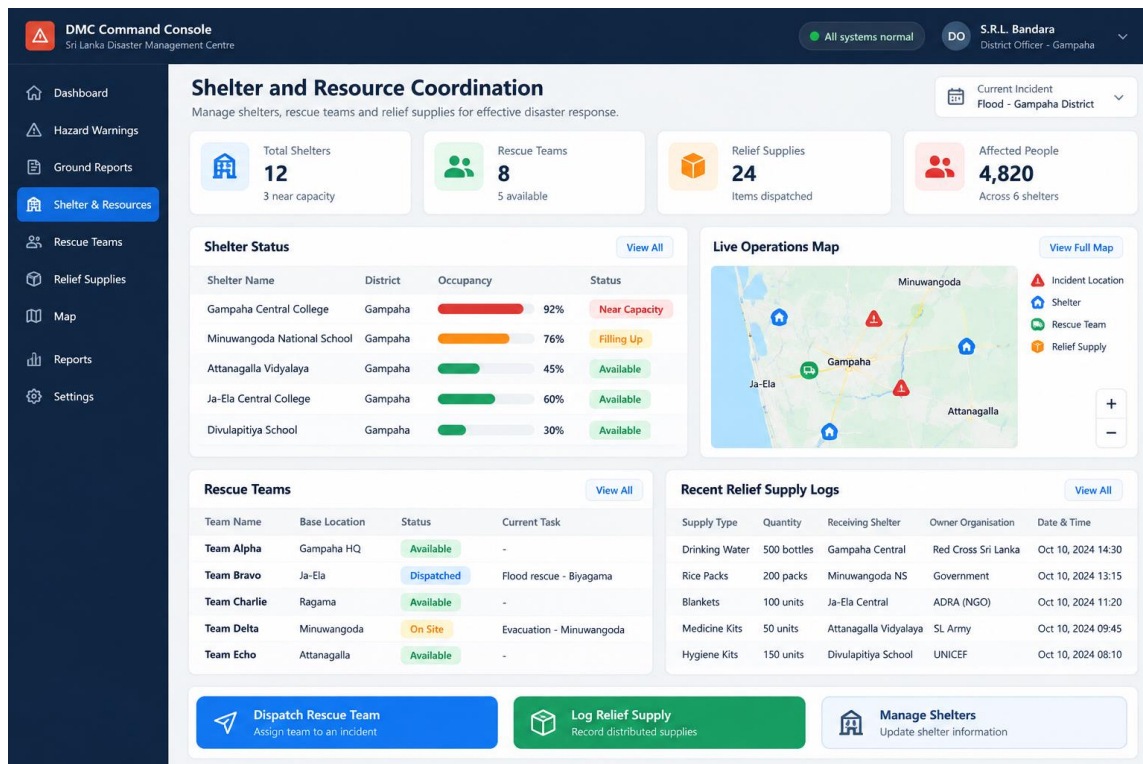


Figure 6: High-fidelity wireframe -Shelter and Resource Dashboard.

6. References

The following existing systems, provided as background reading in the case study, informed the design of this platform:

- Sri Lanka Disaster Aid Management Platform (SL Relief)
- Japan's J-Alert National Warning System
- USA's FEMA Disaster Management Systems
- EU's Copernicus Emergency Management Service

Note on assumptions: the case study brief is intentionally incomplete, ambiguous and vague, as is typical of an industry client brief. Where details were not specified (e.g. exact severity-level names, verification-timeout durations, and the specific fields captured on each screen) the group made reasonable design assumptions consistent with the stated goals of full functionality and high usability under stressful, time-critical conditions.